Nate Cirino

832-833-5202 | cirino.na@northeastern.edu | portfolio | linkedin.com/in/nate-cirino | github.com/nateci

Graduating May 2027 | Available May 2027 for a 2027 start

Education

Northeastern University May 2027 | MGPA: 3.6/4.0
B.S. in Computer Science, Systems Concentration *Boston, MA*

Coursework: Computer Systems, Foundations of Data Science, Algorithms & Data Structures, Computer Networks, Digital Design & Computer Architecture, Object-Oriented Design, Foundations of Software Engineering, Systems Security, Cybersecurity

Experience

Wayfair Jul 2026 – Present
Software Engineering Co-op *Boston, MA*

- Own front-of-house device stack across all Wayfair physical retail brands (AllModern, Birch Lane, Joss & Main, Perigold): e-labels, mobile, and customer/associate-facing POS
- Building a supplier-agnostic e-label API that abstracts 4 heterogeneous suppliers behind one scalable interface, powering new outlet launches
- Shipped POS checkout flows with real-time data sync: in-store scan-to-basket customer switching and live associate-entry customer updates

Wolters Kluwer Jul 2025 – Jun 2026
Software Engineer *Boston, MA*

- Root-caused and overhauled Relay JMS message-broker serialization with json-io across 20+ Java microservices, eliminating production timeouts at scale
- Optimized Expert AI search indexing and filtering across 20+ microservices, improving clinical query relevance at scale
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing workflow
- Integrated Expert AI into CME accrual/redemption microservices, expanding GenAI CDS to 250K+ clinicians
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians

SculptAI Jan 2025 – Aug 2026
Software Architect *New York, NY*

- Maintained 99.99% uptime and high availability across 1–2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Architected Flask backend and retrieval pipeline serving personalized workout recommendations from Bedrock KB
- Evolved retrieval stack from LangChain to Vespa distributed search to zero-ops Bedrock KB, improving relevance and reducing operational overhead
- Led eng org restructure at $1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows

Chevron New Energies May 2024 – Aug 2024
Software Engineer Intern *Houston, TX*

- Built offline data pipelines and RESTful APIs using Azure Data Factory, improving product efficiency by 40%
- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback

Projects

Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio* Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and an LSM-tree storage engine, providing scalable high-performance storage
- Implemented Raft leader election, log replication, and membership changes for stability and high availability; storage layer: memtable, write-ahead log, leveled SSTables
- Added MVCC transaction layer providing snapshot isolation and linearizable reads across multi-node Raft cluster
- Built single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution

Sentinel | *Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker* Apr. 2026 – Present

- Built tooling for production-system stability: an AI SRE agent autonomously triaging incidents via tool-use over Prometheus, Tempo, and Loki telemetry
- Built pgvector RAG over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

NOAA AUV Sonar Detection | *Python, PyTorch, AWS Neuron SDK, SageMaker* April 2025 – July 2025

- Applied ML over large unstructured sensor datasets, training CNN models on thermal spectrograms to detect novel aquatic signatures for NOAA AUVs
- Refined architecture and loss formulation to address class imbalance and sensor noise, improving model detection
- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time limits

Skills & Honors

Languages/Tools: Java, C++, C, Python, Rust, Bash, TypeScript, JavaScript, Linux, PostgreSQL, GraphQL, AWS, Azure, Docker, Kubernetes, Terraform, Jenkins, Buildkite, Groovy

Distributed Systems & Data: Raft consensus, LSM-tree & SSTable compaction, MVCC, write-ahead logging, JMS message brokers, Vespa, pgvector, Azure Data Factory pipelines, Datadog, Prometheus

Frameworks & Libraries: Spring, Flask, FastAPI, PyTorch, TensorFlow, Pandas, Node.js, Express, React, Next.js

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Honors: Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor